Branch Coverage Report

for

Student Attendance Management System (SAMS)

Version: 1.0

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Team 1

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1. Introduction

This report gives the branch coverage of the SAMS source code by the automated test suite. Branch coverage = decision outcomes (each true and false path of every if, for, while and try) taken by at least one test / total decision outcomes. A "partial" branch is a decision where only one of its two paths was taken.

2. Tool and Command

Item	Details
Tool	coverage.py 7.17.0 (with --branch)
Command	cd sams USE_SQLITE=1 coverage run --branch --source=. manage.py test coverage report -m --omit="*/migrations/*,manage.py,sams/*,*/tests.py,*/apps.py,*/__init__.py" coverage html (same --omit)
Excluded	Migrations, Django settings / urls / wsgi (sams/), test files, apps.py, empty __init__.py
Test suite	121 tests, all passing (see Unit Test Report)
Date run	02-11-2026

3. Summary

Branches	Fully covered	Partial	Missed outcomes	Branch coverage
160	150	8	10	93.8%

Before the testing week branch coverage was 83.8% (134 of 160 outcomes). After adding tests for form validation, Excel import rules, the admin forms and the leave decision paths it is 93.8% (150 of 160).

4. Coverage by File

File	Branches	Covered	Partial	Coverage	Missed branches
academics/admin.py	4	3	1	75.0%	23->25
accounts/decorators.py	2	2	0	100.0%	-
accounts/forms.py	2	2	0	100.0%	-
accounts/importer.py	38	38	0	100.0%	-

accounts/models.py	4	4	0	100.0%	-
accounts/views.py	6	5	1	83.3%	21->37
attendance/forms.py	4	4	0	100.0%	-
attendance/utils.py	4	4	0	100.0%	-
attendance/views.py	20	19	1	95.0%	111->131
backup_database.py	2	0	0	0.0%	20->21, 20->23
leaves/forms.py	12	10	2	83.3%	33->47, 38->47
leaves/utils.py	6	6	0	100.0%	-
leaves/views.py	12	12	0	100.0%	-
reports/exports.py	18	16	2	88.9%	56->55, 86->84
reports/forms.py	8	7	1	87.5%	17->19
reports/services.py	8	8	0	100.0%	-
reports/views.py	10	10	0	100.0%	-
TOTAL	160	150	8	93.8%	

Files with no decisions (models with only fields, urls.py, admin registrations) have no branches and are not listed.

5. Code Not Covered and Reason

File	Branch	Path not taken
academics/admin.py	23->25	Class form field other than "students" - the model has no other many-to-many field
accounts/views.py	21->37	Admin opening the empty upload page with GET (only 403 for other roles and POST were tested)
attendance/views.py	111->131	Faculty loads an editable class and leaves without saving (GET path)
leaves/forms.py	33->47	clean() when a date field is already invalid (Django stops earlier with a field error)
leaves/forms.py	38->47	Form created without a student - views always pass the logged in

		student
reports/exports.py	56->55	PDF export loop with a student at or above 75% after a student below 75% in the same file
reports/exports.py	86->84	Same as above for the Excel export
reports/forms.py	17->19	Report form for an admin user (only faculty were tested)
backup_database.py	all	Script not run by the test suite (see Code Coverage Report)

6. Conclusion

93.8% of all decision outcomes are tested. All business rules with a decision (75% threshold BR-4, 3-day edit limit BR-3, role checks NFR-7, duplicate marking REQ-9, leave overlap and date rules REQ-17) are covered on both paths. The missed paths are listed above and are low risk.